

Talha Ghafoor

AI & Machine Learning Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Software Engineering student focused on AI systems, NLP, and cloud-native ML deployment. Experienced in building full-stack AI applications using Python, FastAPI, React, Docker, and AWS. Strong foundation in ML pipeline development, serverless infrastructure, and end-to-end product engineering. Seeking a junior role in AI/ML engineering or full-stack development.

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineering Intern

July 2025 – September 2025

Digital Empowerment Network

- Built a mental health detection model using TF-IDF and Logistic Regression on unstructured text data, achieving approximately 84% F1-score after iterative feature engineering and threshold tuning.
- Developed a heart disease detection pipeline; applied SMOTE to address class imbalance in clinical datasets, improving minority-class recall from approximately 71% to 83%.
- Delivered end-of-internship evaluation reports covering accuracy, precision, recall, and AUC-ROC scores to the supervising engineering team.

AI and Machine Learning Engineering Intern

June 2025 – July 2025

DevelopersHub Corporation

- Trained a CNN-based image classifier achieving approximately 89% validation accuracy through architecture tuning and data augmentation on a dataset of around 6,000 images.
- Cleaned, labeled, and augmented training data to reduce overfitting and improve generalization across unseen test samples.
- Documented model performance across training runs and submitted structured weekly progress reports.

KEY PROJECTS

GenMark – Multi-Modal AI Content Generation Platform

2025 – Present

Final Year Project (FYP) – HITEC University

Python, FastAPI, React, AWS, Docker

- Designed an on-demand AI inference system on AWS using a lazy-loading architecture where models initialize per request, reducing idle compute cost versus always-on deployments.
- Built a serverless routing layer with AWS API Gateway and Lambda to dynamically dispatch requests across text and image generation pipelines.
- Deployed text models on ECS Fargate (CPU); integrated HuggingFace Inference API for image generation (Realistic Vision V5.1) with outputs stored in AWS S3.
- Integrated Qwen for text generation and Stable Diffusion for image generation; containerized all services with Docker and managed artifacts via Amazon ECR.
- Built a React frontend for real-time content generation and preview.

Full-Stack E-Commerce Platform

2025

Next.js, Laravel, Supabase (PostgreSQL), Stripe, AWS S3

- Built a complete e-commerce application covering six flows: product discovery and SEO, storefront browsing, cart and checkout, Stripe payment processing, order fulfillment, and automated postpurchase emails.

- Developed a Next.js frontend with a custom Tailwind CSS design system and a Laravel REST API backend; used Supabase (PostgreSQL) as the primary database and AWS S3 for image storage and CDN delivery.
- Implemented Stripe webhooks for real-time order status synchronization between payment gateway and backend.

NLP Mental Health Classification System2025

Python, scikit-learn, NLP

- Built a TF-IDF and Logistic Regression classifier for early mental health indicator detection, evaluated using precision, recall, and F1-score across multiple class categories.

Fake News Detection System2024

Python, NLP, Streamlit

- Built a fake news classifier using Logistic Regression and TF-IDF; deployed as a Streamlit web application with real-time classification and performance visualizations.

TECHNICAL SKILLS

Core Languages: Python, JavaScript

Familiar With: Java, C++, Dart

AI & Machine Learning: scikit-learn, NLP, Computer Vision, CNN, TF-IDF, HuggingFace Transformers, Stable Diffusion, Qwen, OpenCV, MediaPipe

Web Development: React, PHP ,Laravel, FastAPI, Flask, REST APIs, Tailwind CSS, HTML, CSS

Databases: PostgreSQL (Supabase), MongoDB, MySQL

Cloud & Infrastructure: AWS (ECS Fargate, Lambda, API Gateway, S3, ECR), Oracle Cloud Infrastructure (OCI), Docker

Tools: Git, GitHub, Jupyter Notebook, VS Code

EDUCATION

BS Software Engineering2022 – 2026 (Expected)

HITEC University, Taxila – Department of Computer Science

Relevant Coursework: Machine Learning, Cloud Computing, Web Engineering, Software Quality Assurance, Data Structures & Algorithms, Database Systems

Intermediate – ICS (Computer Science)2020 – 2022

KRL College, Kahuta

Matriculation – Computer Science2018 – 2020

Mumtaz Grammar School, Wazirabad

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate August 2025 Oracle University | Valid until August 2027

Certified Python & Machine Learning2025

Mars Computer Academy, Islamabad